AI-Based Diabetes Prediction and Risk Factor Analysis System

1. Project Overview

This project aims to develop a Machine Learning-based system for predicting the likelihood of diabetes based on an individual's health, demographic, and lifestyle-related characteristics.

The project will focus on developing a complete Machine Learning pipeline, starting from data understanding and preprocessing, followed by exploratory data analysis, feature selection, model training, evaluation, and comparison.

In addition to predicting diabetes, the project will investigate the relationship between different health and lifestyle factors and diabetes in order to identify the factors that have the strongest association with the prediction.

The project is intended as an educational and research-oriented Machine Learning system and is not designed to provide medical diagnosis or replace professional medical evaluation.

2. Problem Statement

Diabetes is a common chronic disease influenced by a combination of demographic, health, and lifestyle factors. Machine Learning can be used to analyze these factors and identify patterns associated with diabetes.

The main challenge addressed by this project is to develop a reliable classification model that can distinguish between individuals with and without diabetes based on available health-related information, while also providing meaningful analysis of the factors associated with the prediction.

3. Project Objectives

The main objectives of the project are to:

- Collect and prepare a suitable diabetes-related dataset.
- Perform Exploratory Data Analysis (EDA) to understand the characteristics and distribution of the data.
- Identify missing values, inconsistencies, outliers, and class imbalance where applicable.
- Apply appropriate data preprocessing techniques.
- Analyze and select relevant features for model development.
- Train multiple Machine Learning classification models.
- Compare the performance of the developed models.
- Evaluate the models using appropriate classification metrics.
- Analyze the relationship between health and lifestyle factors and diabetes.
- Select the most suitable model based on its overall performance.

4. Dataset

The proposed dataset is the **Diabetes Health Indicators Dataset**, which is based on health and lifestyle information collected through the Behavioral Risk Factor Surveillance System (BRFSS).

The dataset contains a large number of records and multiple health, demographic, and lifestyle-related features.

Potential features include:

- Age
- BMI
- High blood pressure

- High cholesterol
- Physical activity
- Smoking-related indicators
- General health
- Physical health
- Mental health
- Education
- Income
- Other relevant health indicators

The target variable will represent the diabetes-related classification provided by the selected dataset. Before model development, the dataset will be carefully examined to ensure that the selected features and target variable are suitable for the project's objectives.

5. Methodology

Step 1: Data Collection and Understanding

The dataset will be examined to understand its structure, features, target variable, data types, and class distribution.

Step 2: Exploratory Data Analysis

EDA will be performed to identify important patterns and relationships within the dataset. This may include:

- Distribution analysis
- Correlation analysis
- Class distribution
- Feature relationships
- Identification of unusual or inconsistent values

Step 3: Data Preprocessing

The data will be prepared for Machine Learning by applying appropriate preprocessing techniques, such as:

- Handling missing values
- Encoding categorical variables when required
- Feature scaling when required
- Handling class imbalance where necessary
- Removing or addressing irrelevant information

Step 4: Feature Selection and Engineering

Relevant features will be analyzed and selected based on their relationship with the target variable. Feature engineering techniques may also be applied when appropriate to improve model performance.

Step 5: Model Development

Three Machine Learning classification algorithms will be developed and compared:

1. Logistic Regression

Used as a strong and interpretable baseline classification model.

2. Decision Tree

Used to model relationships between health-related features and diabetes through a tree-based decision structure.

3. Random Forest

Used as an ensemble model that combines multiple decision trees and can capture more complex relationships within the data.

Step 6: Model Evaluation

The three models will be evaluated and compared using several performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Special attention will be given to **Recall and False Negatives**, since incorrectly classifying an individual with diabetes as negative can be particularly important in a healthcare-related prediction problem.

Step 7: Model Comparison

The results of the three models will be compared to determine which algorithm provides the most suitable overall performance for the selected dataset. The comparison will consider not only accuracy, but also precision, recall, F1-score, ROC-AUC, and the confusion matrix.

6. Proposed System Workflow

The overall workflow of the project will be:

Dataset → Data Cleaning → EDA → Preprocessing → Feature Selection → Model Training → Model Evaluation → Model Comparison → Final Diabetes Prediction

7. Expected Results

The expected outcome of the project is a Machine Learning model capable of predicting the diabetes-related class of an individual based on the available health and lifestyle features.

The project is also expected to provide:

- A clear understanding of the dataset and its characteristics.
- Identification of important diabetes-related features.
- A comparison between three different classification algorithms.
- A detailed evaluation of model performance.
- Selection of the best-performing model for the given dataset.
- Analysis of the strengths and limitations of the developed models.

8. Technologies and Tools

The project will primarily use:

- **Python**
- **Pandas** for data manipulation
- **NumPy** for numerical operations
- **Matplotlib** and **Seaborn** for data visualization
- **Scikit-learn** for Machine Learning models and evaluation
- **Jupyter Notebook** for experimentation and analysis

9. Project Scope

The project will focus on developing and evaluating Machine Learning classification models for diabetes prediction using existing health-related data.

The project will not attempt to provide medical diagnosis or guarantee whether an individual will develop diabetes in the future. Instead, the model will estimate the likelihood of diabetes based on patterns learned from the available dataset.

The final results will be interpreted within the limitations of the dataset and the selected Machine Learning methods.

10. Conclusion

This project will demonstrate the application of Machine Learning to a real-world healthcare-related problem. It will cover the complete Machine Learning workflow, including data analysis, preprocessing, feature selection, model development, evaluation, and comparison.

By comparing Logistic Regression, Decision Tree, and Random Forest, the project will provide practical insight into how different Machine Learning approaches perform on diabetes-related data and which approach is most suitable for the selected dataset.